Artificial Intelligence Basics

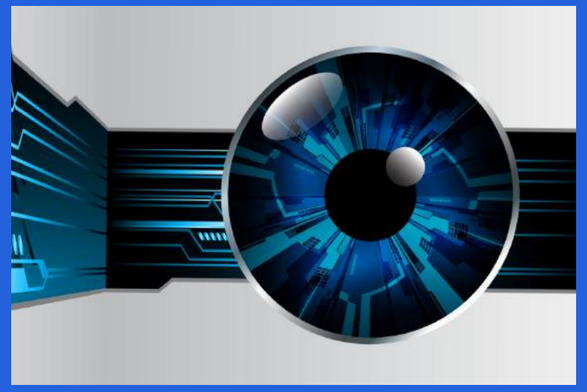
What is Artificial Intelligence?

- AI means machines acting smart like humans.
- It can learn, think, and solve problems.
- AI copies how humans use their brains.
- Example: When your phone suggests words — that’s AI!

Main Abilities of Artificial Intelligence

LEARNING

AI systems improve through data analysis.



REASONING

AI can make decisions based on logic.

PERCEPTION

AI understands images and sounds

LANGUAGE PROCESSING

AI understands and generates human language.

Types of AI

NARROW AI

Narrow AI focuses on **specific tasks** and is commonly used in applications like voice assistants and recommendation systems.

GENERAL AI

General AI aims to replicate human-level thinking, with an ability to learn and apply knowledge across various domains. (not real yet)

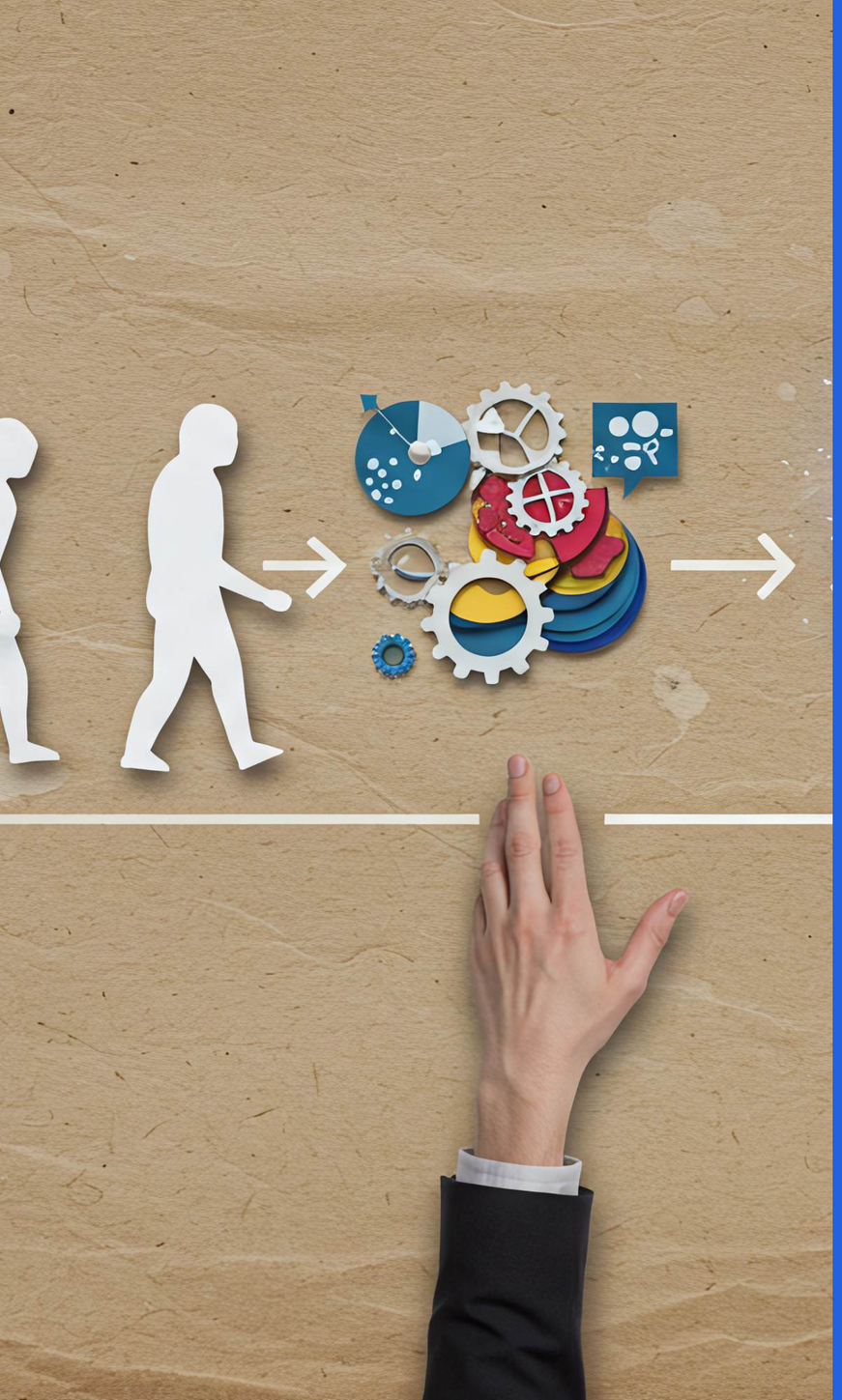
SUPERINTELLIGENT AI

Superintelligent AI refers to a future type of AI that is smarter than humans and can solve problems in ways we cannot fully understand. (only theory)

Exploring Weak AI: Real-World Examples

NARROW AI IN ACTION

Weak AI, such as **Siri, Alexa, and ChatGPT**, excels in specific tasks, providing assistance and enhancing user experiences across various platforms.



General AI: The Future of Intelligence

General AI aims to achieve **human-like thinking**, allowing machines to learn and adapt. It remains a theoretical pursuit, not yet realized.

Superintelligent AI: Capabilities Beyond Human Reach

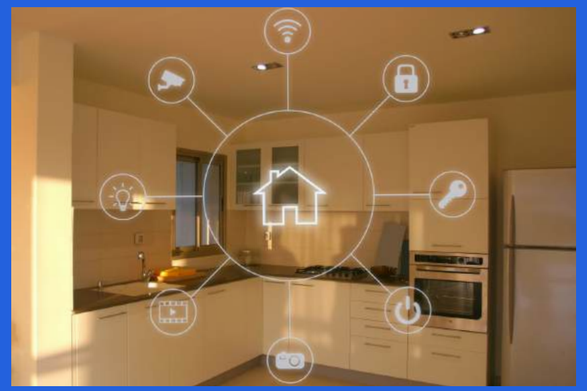
THEORETICAL FUTURE POTENTIAL

Superintelligent AI is a future idea where AI becomes smarter than humans, which raises big questions about technology and ethics

AI Applications in Our Daily Lives

STREAMING

AI recommends shows based on preferences.



SMART HOMES

AI controls lighting and security systems.

VOICE ASSISTANTS

AI helps perform tasks through voice commands.

TRANSPORTATION

AI optimizes routes for self-driving cars.

Common AI Myths

- AI has no feelings — it doesn’t get happy or sad.
- AI won’t take every job, it creates new ones too.
- AI can make mistakes if data is wrong.
- AI can’t think by itself.
- AI won’t take over the world like in movies!

Core AI Concepts

ALGORITHMS

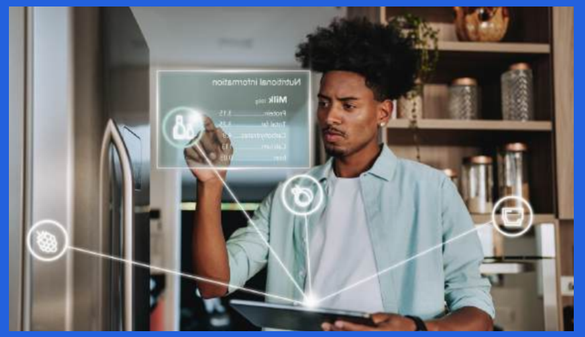
Rules that guide AI decision-making processes.

MODELS

Frameworks built to predict outcomes based on data.

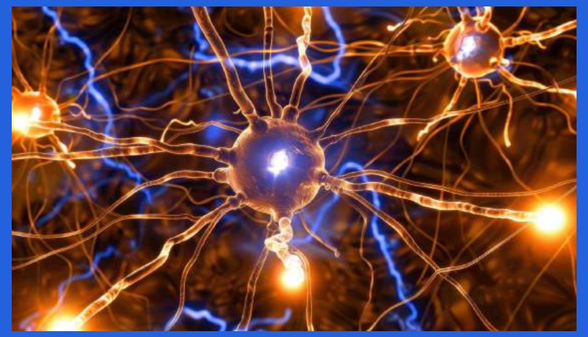
TRAINING DATA

Data used to improve AI's performance and accuracy.



INFERENCE

AI's ability to derive conclusions from data.



NEURAL NETWORKS

Systems modeled after the human brain for learning.

NLP

Techniques for processing and analyzing human language.

How AI Works (Core Principles)

LOGIC & RULES

AI relies on fundamental **logic and rules** for tasks.

PROBABILITY

Probability helps AI predict outcomes and decisions effectively.

LEARNING

Learning enables AI to improve through **experience and data**. to get better over time

PATTERN RECOGNITION

Find trends in data

TRIAL & ERROR

Learn by trying again and again

Summary

- AI means smart machines that learn and think.
- It already helps us every day.
- It can make mistakes — it’s not perfect.
- AI’s goal: help humans, not replace them.